

Purella Saikrishna

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Objective

Data Analyst with nearly 3 years of experience in data analysis, statistical techniques, and machine learning. Passionate about leveraging AI/ML techniques to drive business solutions across industries. Proficient in SQL, Python, data visualization, and statistical analysis.

Professional Work Experience

Optum - Data Analyst, Hyderabad

Sep 2022 – June 2025

Role and Responsibilities:

- Analyzed and interpreted structured and semi-structured healthcare datasets to drive actionable insights and improve operational performance, while developing rule-based logic and predictive models to optimize healthcare processes and detect errors efficiently.
- Designed and maintained interactive dashboards using Power BI to provide real-time reporting and trend visualization for stakeholders.
- Executed data extraction, transformation, and loading operations using SQL and Python across multiple healthcare data sources.
- Collaborated cross-functionally with business, operations, and quality teams to define KPIs and establish robust reporting frameworks.
- Prepared and implemented data validation frameworks ensuring data integrity and compliance with healthcare regulations.

Skills: SQL, Python, Power BI, Tableau, Excel, Healthcare Data Analytics, Predictive Modeling, ETL, Data Visualization.

Internship and freelance ML Engineer

For AITAR Pvt. Ltd., Hyderabad for Geopredict GmbH, Germany

Jan 2022 - July 2022

Project: SEED, ARTIGROW - utilizing CLIMFOR modeling and forecasting technology for Energy and Agriculture.

Role and Responsibilities:

- Collected and aggregated historical data from various sources, including the Copernicus website, for climate change and solar energy forecasting.
- Designed and managed SQL databases to store, retrieve, and analyze large datasets efficiently, ensuring seamless data accessibility for machine learning workflows.
- Preprocessed and cleaned data, performed feature engineering, and validated model performance in collaboration with team members.
- Contributed to the implementation and optimization of machine learning algorithms, including decision trees, clustering, and neural networks, to enhance forecasting accuracy and efficiency.
- Applied a range of techniques such as classification, regression, summarization, and dependency modeling to support project objectives.

Personal Projects

Claim Denial Prediction System:

- Developed a machine learning pipeline using Logistic Regression to predict healthcare claim denial risk based on patient demographics, provider type, diagnosis codes, and claim history.
- Engineered features and automated data preprocessing with SQL and Python, including handling missing values, categorical encoding, and anomaly detection.
- Designed interactive Power BI dashboards to visualize denial trends, top denial reasons, and provider performance for business stakeholders.
- **Technologies:** Python (Pandas, Scikit-learn), SQL, Power BI

AI-Powered Desktop Automation Agent:

- Designed and deployed a cloud-native desktop automation system on Google Cloud Platform using microservices architecture with Cloud Run, Cloud Functions, and Firestore.
- Developed a Windows desktop agent with screen capture, input control, and WebSocket communication modules for real-time client-server interactions.
- Integrated OpenAI GPT-4 Vision for screen understanding and GPT-4 for AI-driven task planning to automate complex business workflows.
- Architected scalable data pipelines using Cloud Storage and Pub/Sub for task queuing, state management, and asynchronous processing.
- **Technologies:** Python, Google Cloud Platform (Cloud Run, Cloud Functions, Firestore, Cloud Storage, Pub/Sub), OpenAI API (GPT-4 Vision, GPT-4), WebSocket.

Skills

- **Programming & OOP:** Python (OOP, API development, testing, documentation), SQL
- **Machine Learning:** Regression, Classification, Clustering, Random Forests, Gradient Boosting, SVM, Ensemble methods
- **Deep Learning:** TensorFlow, Keras, PyTorch (CNNs, RNNs, Transformers)
- **NLP:** Text preprocessing, Named Entity Recognition, Sentiment Analysis, Hugging Face Transformers, GPT/BERT models
- **Data Engineering & ETL:** Data pipelines, large-scale data handling, PySpark, SQL-based ETL processes
- **MLOps & Deployment:** Model versioning, monitoring, retraining, API development (FastAPI, Flask), Docker, Kubernetes
- **Cloud Platforms:** Google Cloud (Vertex AI, Cloud Run, Functions, Firestore, Pub/Sub)
- **Big Data Tools:** PySpark, Hadoop (basics)
- **Visualization:** Tableau, Power BI
- **Advanced Analytics:** Time series forecasting, anomaly detection, recommendation systems,
- **Version Control & CI/CD:** Git, automated testing, deployment pipelines
- **Databases:** SQL Server, MySQL, NoSQL, Vector databases

Education

- Bachelor of Science in Computer Science

2021